

Holography for people with no time

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Abstract

We study the gravitational description of extremal supersymmetric black holes. We point out that the AdS_2 near horizon geometry can be used to compute interesting observables, such as correlation functions of operators. In this limit, the Hamiltonian is zero and correlation functions are time independent. We discuss some possible implications for the gravity description of black hole microstates. We also compare with numerical results in a supersymmetric version of SYK. These results can also be interpreted as providing a construction of wormholes joining two extremal black holes. This is the short version of a longer and more technical companion paper [\[1\]](#).

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1 Introduction

Charged extremal black holes are very interesting objects. They seem to defy the third law of thermodynamics since they have non-zero entropy at zero temperature. In addition, their geometries develop a seemingly infinite throat with an $AdS_2 \times S^2$ (or $AdS_2 \times (\text{something})$) geometry. These were the type of black holes for which the Bekenstein-Hawking entropy was first matched with a microscopic counting [2].

It was recently understood that there is a gravitational mode whose quantum fluctuations become large as we take the low energy limit [3, 4, 5, 6]. Fortunately, we can solve exactly the quantum mechanics of this single mode [7, 8]. This modifies the naive classical gravity results at low temperatures. For the non-supersymmetric case, this implies that the density of states vanishes at low energies [9, 10, 8]. On the other hand, with $\mathcal{N} = 2$, or $\mathcal{N} = 4$, supersymmetry the density of states has a gap above zero and there is a large degeneracy at exactly zero energy above extremality [10, 11]. The $\mathcal{N} = 2$ arises for BPS black holes in AdS_5 [12], while the $\mathcal{N} = 4$ one describes supersymmetric black holes in $D = 4, 5$ flat space with non-zero horizon area [11].

This energy gap means that by taking a low energy limit we can clearly restrict to the ground states. In other words, we have a decoupling limit which allows us to isolate the ground states. The extremal entropy is a well studied observable in this limit, starting from [2] and including very detailed matches, as in [13].

Here we describe another set of observables which consist of correlation functions of certain operators. We consider “simple” operators that correspond to bulk fields located near the AdS_2 boundary, or near the region where the AdS_2 throat opens up into a higher dimensional spacetime. These correlators are given by Witten diagrams in AdS_2 which are dressed by the quantum dynamics of the boundary graviton mode, see figure 5. At low energies these correlators develop a certain universal time dependence that depends only on properties of the boundary graviton mode. For the supersymmetric case, the situation is particularly simple: they are completely time independent at very large times. These constant values depend on the details of the bulk AdS_2 theory.

In a dual quantum mechanical theory we can interpret these correlators as

$$\text{Tr}[\hat{O}_1 \hat{O}_2 \cdots \hat{O}_n], \quad \hat{O}_i = P O_i P, \quad P = \lim_{u \rightarrow \infty} e^{-uH} \quad (1)$$

where O_i are the simple operators and P is the projector onto zero energy states. As an example of a quantum mechanical theory with these properties we study the supersymmetric SYK model introduced in [14]. For a case with an Einstein gravity dual, we can consider the supersymmetric black hole in $AdS_5 \times S^5$ [15], whose low energy boundary gravity mode has $\mathcal{N} = 2$ supersymmetry [12], which is the case we analyze in detail. However, we expect that other supersymmetric black holes, which have a $\mathcal{N} = 4$ boundary graviton mode [11], would have similar properties.

The projector operator P in (1) can be viewed as describing the zero temperature limit of the thermofield double state. More explicitly, when we write e^{-uH} on the RHS of (1), we may view this as an operator acting on the Hilbert space of the one-sided quantum mechanical dual of the gravitational system, or we may view it as defining a two-sided state, e.g., the thermofield double with inverse temperature $\beta = 2u$. Taking $u \rightarrow \infty$ corresponds to the zero temperature limit of the usual two sided black hole. According to the classical solution, the length of the wormhole goes to infinity as $T \rightarrow 0$. However quantum effects kick in at temperatures of order the energy gap, and we find that its length remains finite at zero temperature. More precisely, the zero temperature state has a normalizable wavefunction which peaks at a finite length which is logarithmic in the extremal entropy, $\log S_e$.

This gives an explicit construction of a supersymmetry preserving wormhole joining two supersymmetric black holes. We can construct a large family of such wormholes by adding operators, see figure 6. We discuss how the addition of matter changes the length, making it larger. Among this family, the empty wormhole has the largest entanglement entropy, equal to the extremal entropy. The others have a smaller one.

Although the Hamiltonian is zero from the boundary¹ point of view, the bulk matter propagates and moves subject to the bulk time. So, we have a clearly emergent bulk time from a dual boundary theory with no time. In other words, an observer deep in the bulk still experiences time, despite the fact that there is no time on the boundary theory in this limit.

In this paper we summarize results from its technically heavier (super) partner [1]. We also discuss some of the conceptual implications.

This paper is organized as follows. In section 2, we consider the dynamics of the boundary graviton mode with $\mathcal{N} = 2$ supersymmetry. We discuss some implications for the description of the microstates. In section 3, we report on a numerical computation of similar correlators in $\mathcal{N} = 2$ SYK. As expected, we find agreement with the analytic computations described in the previous section, since they are governed by the effective theory. We end with further discussion in section 4.

2 The zero energy limit of the $\mathcal{N} = 2$ JT gravity theory

2.1 Density of states and the gap

By taking the extremal black hole limit, we naively expect to get a scale invariant system. But scale invariance is not compatible with the discreteness of the spectrum. One exception is if all states are precisely degenerate, so that the Hamiltonian is zero. Then the theory is not only scale invariant but also completely time independent. It becomes fully time reparametrization invariant.

In supersymmetric cases, with $\mathcal{N} = 2, 4$ supersymmetry, this is precisely what happens. One can compute the density of states using the cigar (or disk) topology and we obtain an answer with the qualitative features in figure 1 [10, 11]. There is a continuum separated by a gap from a

¹By boundary, we mean the boundary of the $NAdS_2$ region, or the putative quantum mechanical dual of the region.

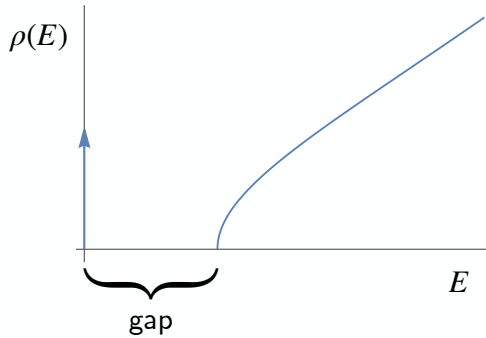


Figure 1: The density of states for the $\mathcal{N} = 2$ superSchwarzian theory in the zero R-charge sector, from [10]. We see the presence of a gap, and a delta function contribution at zero which gives the extremal entropy. $E_{\text{gap}} = \frac{1}{32C}$, with C defined in (2).

delta function containing a large number of degenerate zero energy states². In many cases, index arguments indicate that this degeneracy should remain in the exact theory [2].

Let us comment on the energy scale that sets the gap. As we go to low energies, there is a particular gravitational mode that becomes strongly coupled. This mode can be described in terms of a reparametrization between the *AdS* time coordinate f and the asymptotic, or boundary, time t [17, 5, 4, 6, 14]

$$I = -C \int \{f(t), t\} + \text{SUSY partners} , \quad \text{with} \quad \{f, t\} = \frac{f'''}{f'} - \frac{3}{2} \left(\frac{f''}{f'} \right)^2 \quad (2)$$

where the supersymmetric partners include fermions plus a second scalar mode we will discuss later. The coefficient C has units of time and its inverse sets the scale of the gap in figure 1. When we consider a charged black hole in flat space this coefficient is $C \sim S_e r_e$, where r_e is the extremal radius of the black hole and S_e is its extremal entropy. Notice that this is a very long time for a large black hole. C also sets the time scale at which the action (2) becomes strongly coupled. We are asserting that for time scales larger than C , the correlators become constant, or time independent.

2.2 The two point functions at long distances

In this section, we sketch why the two point function has a constant value at long times

$$\langle O(u)O(0) \rangle_\beta \rightarrow \text{constant} , \quad \text{for} \quad u, \beta, \beta - u \gg 1 \quad (3)$$

These correlation functions can be computed as follows. We can view them as the matrix element of an operator between two wormhole states, one that has been generated via euclidean time u and another with time u' , see figure 2. We will denote such wormhole states as $|\text{TFD}(u)\rangle$. We then want to compute

$$\langle \text{TFD}(u') | e^{-\Delta \ell} | \text{TFD}(u) \rangle \quad (4)$$

where Δ is the conformal dimension of the operator O . Each of these thermofield double states is a quantum superposition of wormhole states with different lengths [7, 8]. More precisely, in the

²The clear gap is a feature of the semiclassical analysis. Once we include e^{-S_0} corrections, due to other topologies, we expect that the eigenvalue distribution will become smooth and there could be a non perturbatively small probability of finding an energy level in the gap region. See figure 13 in [16] for an example. We ignore such corrections here. We will show that in the SYK model for small N the gap is typically clearly present.

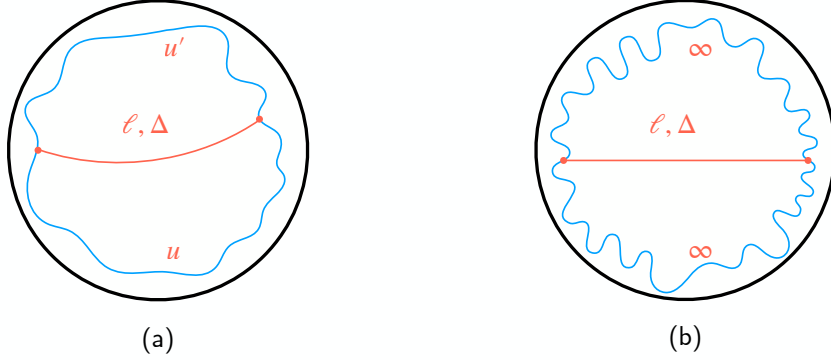


Figure 2: Diagrams for two point function. (a) The bottom propagator generates the thermofield double state $|\text{TFD}(u)\rangle$ and the top one generates a similar one with $u \rightarrow u'$. The correlator involves a geodesic going between the two boundary points whose renormalized length is ℓ and the associated conformal dimension is Δ . (b) The correlator in the $u = u' = \infty$ limit. The boundary has large fluctuations but the distance between the two operator insertions remains finite.

supersymmetric case, the length also has further fermionic and bosonic partners that arise from the action of the supercharges [18]. These variables are governed by a supersymmetric quantum mechanics. In the $\mathcal{N} = 2$ case this action has the form

$$I = \int du \left[\frac{1}{4} \dot{\ell}^2 + \dot{a}^2 + i\bar{\psi}_r \dot{\psi}_r + i\bar{\psi}_l \dot{\psi}_l + \bar{\psi}_l \psi_r e^{-\ell/2 - ia} + \psi_l \bar{\psi}_r e^{-\ell/2 + ia} + e^{-\ell} \right] \quad (5)$$

where we have set units such that $C = 1/2$, or equivalently defined u so that is related to t by

$$t = 2Cu \quad (6)$$

The field a can be viewed as arising from a Wilson line of a $U(1)$ gauge field in the bulk and is related to a boundary $U(1)_R$ symmetry. This field is periodic with a period $a \sim a + 2\pi$, implying that the R charges are integer quantized. It is also possible to make the period larger $a \sim a + 2\pi\hat{q}$ with integer $\hat{q}$, and we will indeed consider this in section 3. To keep the discussion simple, we set $\hat{q} = 1$ for now.

The Lagrangian (5) actually has four supersymmetries. The reason is that we have two from the left boundary and two from the right boundary. Each of those two anticommutes to the same Hamiltonian. One can use these four supersymmetries to determine the form of the Lagrangian by demanding that the potential behaves like $e^{-\ell}$ at large negative ℓ ³.

An important feature of the Lagrangian (5) is that there are two potential terms, a repulsive potential $e^{-\ell}$ and a Yukawa term involving $e^{-\ell/2}$. We can think of the fermions as two qubits. Depending on the state of these qubits the Yukawa term could lead to a positive, zero or negative potential for ℓ . When this $e^{-\ell/2}$ term is negative we can have bound states, see figure 3. It turns out that there is only one bound state, with exactly zero energy and also zero R charge. This is due to the precise relative coefficients of the two terms in the potential, which are fixed by supersymmetry. This state preserves both the left and the right boundary supersymmetries, so it can be viewed as a supersymmetric wormhole.

³The fact that the potential agrees at large ℓ with the $\mathcal{N} = 0$ Schwarzian is required since the classical solutions of the $\mathcal{N} = 0$ Schwarzian are also solutions of the $\mathcal{N} = 2$ Schwarzian with $a = 0$ and all fermions set to zero. By the way, note that the classical solutions in Euclidean signature are circles in H_2 . Since the two sides of the circle meet at a point after proper time $\beta/4$, the renormalized distance $\ell \rightarrow -\infty$ after a time $\beta/4$.

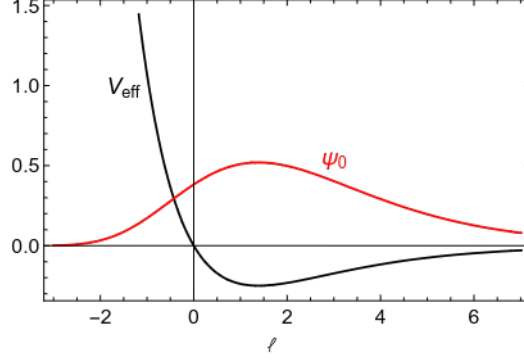


Figure 3: In black, we plot the effective potential when the term involving the fermions has an effectively negative coefficient. It leads to a single normalizable state, which is shown in red.

In this zero energy state the wavefunction has the following ℓ dependence, see figure 3,

$$\langle \ell | 0 \rangle \propto e^{-\ell/2} \exp\left(-2e^{-\ell/2}\right) \times (\text{fermions}) \quad (7)$$

This zero energy state, $|0\rangle$, is unit normalized and it appears in the expansion of the thermofield double as

$$|\text{TFD}(u)\rangle = e^{\frac{S_0}{2}} |0\rangle + \dots \quad (8)$$

where the dots indicate terms that decay as $u \rightarrow \infty$. The zero energy partition function is given by

$$Z = Z(\beta = \infty) = e^{S_0} \quad (9)$$

and it sets the number of ground states. Then the long distance two point function has the form [1]

$$\lim_{u', u \rightarrow \infty} \langle \text{TFD}(u') | e^{-\Delta \ell} | \text{TFD}(u) \rangle = e^{S_0} \langle 0 | e^{-\Delta \ell} | 0 \rangle = e^{S_0} \frac{\Gamma(1 + 2\Delta)}{2^{4\Delta}} \quad (10)$$

where the precise function of Δ in the right hand side requires a calculation using (7). But the fact that it is an order one number for an order one value of Δ does not require any calculation, since we have already said that $|0\rangle$ has a wavefunction localized at a value $\ell \sim 0$, (7). Furthermore, we can calculate the expectation value of ℓ as

$$\langle \ell \rangle = -\partial_{\Delta} \log \langle 2\text{pt} \rangle|_{\Delta=0} = 2(\gamma_E + \log 4) \sim o(1) \quad (11)$$

which is indeed finite and of order one. In addition, since $H = 0$, this distance does not grow in time, in contrast with the behavior at finite temperature.

In order to properly interpret these correlators (10), we need to understand how the operators were normalized. First, the distance variable ℓ has been defined with a subtraction from the real distance $d = \ell - 2 \log \epsilon$, where ϵ is a time cutoff corresponding to the point where the AdS_2 joins flat space. Here both distances are in radius of AdS_2 units. For a Reissner Nordström black hole, $\epsilon \sim r_e/(2C) \sim \frac{1}{S_e}$.

In (10), we have normalized the operators so that their short distance expression is [1]

$$\text{Tr} \left[e^{-(\beta-u)H} O e^{-uH} O \right] = \langle \text{TFD}(\beta-u) | e^{-\Delta \ell} | \text{TFD}(u) \rangle \sim \frac{1}{u^{2\Delta}} Z(\beta), \quad \text{for } u \ll 1 \quad (12)$$

where the $1/u^{2\Delta}$ dependence is set by the conformal limit in AdS_2 , which is a good approximation at relatively short distances, where the Schwarzian mode is weakly coupled. The overall coefficient in (12) sets the normalization.

It is useful to express the answer in terms of an operator W whose two point function is normalized to be of order one in the boundary of AdS_2 . More precisely we set $\langle WW \rangle \sim \left(\frac{r_e}{t}\right)^{2\Delta}$ in terms of the Schwarzschild time t . Recalling (6) we obtain

$$\langle \hat{W}\hat{W} \rangle_\infty = e^{S_0} \left(\frac{r_e}{2C}\right)^{2\Delta} \frac{\Gamma(1+2\Delta)}{2^{4\Delta}} \propto e^{S_0} \frac{1}{S_e^{2\Delta}} \frac{\Gamma(1+2\Delta)}{2^{4\Delta}} \quad (13)$$

The hat notation means that we have evolved over a very long Euclidean time so as to project to the ground states, as in (1).

The factor of e^{S_0} disappears once we divide by the partition function (9). An interesting point about (13) is the power of r_e/C . This is setting the typical proper distance between the two boundaries (in units of the radius of AdS_2)

$$d = \ell + 2 \log \frac{1}{\epsilon} \sim 2 \log \frac{C}{r_e} \sim 2 \log S_e \quad (14)$$

since $\ell \sim o(1)$ for the ground state (11). This should be compared to the naive classical expression

$$d \sim 2 \log \frac{\beta_t}{r_e}, \quad \text{classical} \quad (15)$$

which diverges⁴ as $\beta_t \rightarrow \infty$. This shows that as β_t reaches $S_e r_e \sim 1/E_{\text{gap}}$, the wormhole stops growing. Notice that we are *not* talking about the growth of the wormhole in time, but the growth as we take β larger. It is also true that the wormhole does not grow in time, since $H = 0$ for the ground states. At zero temperature, the system is time independent and the wormhole length stays fixed at (14). Both of these features are in stark contrast with the behavior for the non-supersymmetric case. In that case the typical distance grows without bound either as $\beta \rightarrow 0$ or as time progresses. In the $\mathcal{N} = 0$ case, it has been argued that contributions from non-trivial topologies cause the distance to stop growing at values that are exponentially large in the entropy [19, 20]. Here we see a much smaller value (14), already in the disk approximation.

2.3 The cylinder two point function in the probe approximation

We can also compute the two point function on the cylinder in the probe approximation⁵. Here we have one operator at one end of the cylinder and the other at the other end. We need to sum over all the states of the empty wormhole, which reduces to a sum over all the states of the Liouville-like theory (5) [21]. At low energies, we can concentrate on the zero energy state contribution which gives simply

$$\langle 2 \text{pt} \rangle_{\text{cyl,probe}} = \langle 0 | e^{-\Delta \ell} | 0 \rangle \quad (16)$$

which is the same as what we got for the two point function on the disk up to the number of ground states, or factor of e^{S_0} . We will explain this “coincidence” later (around (27), (28)).

2.4 n -point correlation functions

In the $\mathcal{N} = 0$ case, we can compute correlation functions by “dressing” the rigid AdS_2 correlators with propagators of the boundary particles [7]. In our $\mathcal{N} = 2$ case we can do the same. Again, when we go to long times, we can get the propagator for zero energy states. This is a function of two bulk points (and their superpartners), which we work it out in detail in [1].

⁴The t index in β_t is the inverse temperature in Schwarzschild time t , $t \sim t + \beta_t$, see (6).

⁵We are neglecting loops of particles wrapping the wormhole. This is reasonable when the typical size of the wormhole is large. We expect that this is the case for large Δ . Note that these extra loops could lead to a divergence in the integration over the cylinder size, from the very thin cylinder region. We are assuming this gets cured somehow.

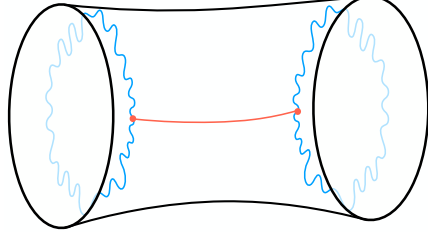


Figure 4: The cylinder diagram in the probe approximation. The states of the empty wormhole propagate along the cylinder.

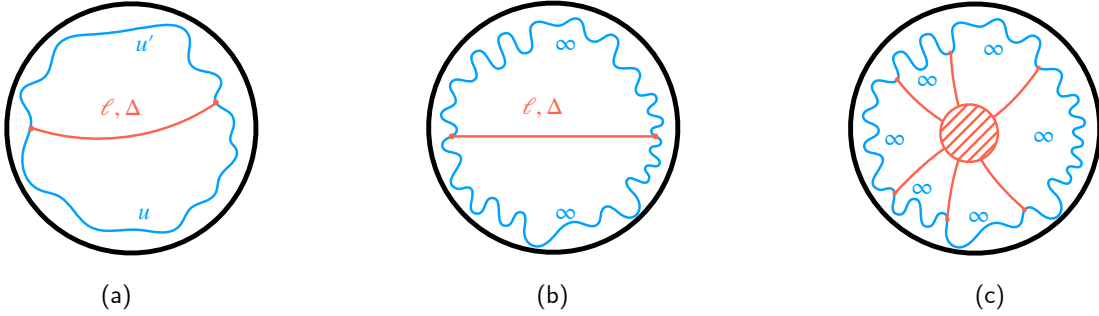


Figure 5: General correlators are built from AdS Witten diagrams, in red, dressed by boundary graviton propagators, in blue.

The correlator is constructed from two elements. First we need the correlator of bulk fields near the boundary of AdS_2 . We take the bulk matter to also be supersymmetric so that we may write their correlators in terms of bulk superfields. This turns out to have the form:

$$\prod_i e^{-\Delta_i \rho_i} \langle O_1(x_1, \theta_{1-}, \bar{\theta}_{2-}) \cdots O_1(x_1, \theta_{1-}, \bar{\theta}_{2-}) \rangle \quad (17)$$

where θ_- and $\bar{\theta}_-$ are Grassmann variables which implement the $\mathcal{N} = 2$ supersymmetry at the AdS_2 boundary. These correlators are computed by starting with the bulk fields in AdS_2 , with no gravity, and then taking them near the boundary.

The second element is the zero energy boundary propagator $P(1, 2)$ which is a function of two points whose coordinates are

$$(x, \rho, a, \theta_-, \bar{\theta}_-, \chi) \quad (18)$$

where χ is an extra Grassmann coordinate that we need to properly describe the Schwarzian degrees of freedom and a keeps track of the R symmetry properties. See [1] for the explicit form of the propagator; it is obtained by demanding that the appropriate supercharges of the super-Schwarzian theory vanish. Importantly, $P(1, 2)$ is independent of the boundary time.

Then the final expression of any correlator at zero energies, or very long times, is given by

$$\begin{aligned} \langle \hat{O}_1 \cdots \hat{O}_n \rangle &= \pi e^{S_0} \int \frac{\prod_i d\mu_i}{\text{Vol}(OSp(2|2))} P(1, 2) P(2, 3) \cdots P(n, 1) \\ &\times \prod_i e^{-\Delta_i \rho_i} \langle O_1(x_1, \theta_{1-}, \bar{\theta}_{2-}) \cdots O_1(x_1, \theta_{1-}, \bar{\theta}_{2-}) \rangle \end{aligned} \quad (19)$$

These correlators do not depend on any boundary times since $H = 0$. But they can depend on the order, though they have a cyclic symmetry, consistent with their holographic interpretation in

(1). The $\int d\mu_i$ integrals are over the variables (18) for each point. The denominator in the measure factor arises because we are gauging the overall $OSp(2|2) = SU(1,1|1)$ symmetry of the integrand.

It turns out that the propagator contains a function of the proper distance that is the same as the wave function we already encountered in (7). This essentially implies that the propagator decays at large proper distances⁶. This implies that the correlators (19) are finite. Another feature of these correlators is that the time ordered and the out of time order correlators are of the same order, at least for low values of Δ .

The time independence of (19) implies that the theory becomes “topological” at zero energies. Of course, topological in one dimension just means that the Hamiltonian is zero and the correlators are independent of time, though they can depend on the ordering. Though the theory is topological in this sense, the operators that we consider are operators that are defined by the higher energy theory. They are simple operators in the higher energy theory that are projected onto zero energy states by performing a large Euclidean time evolution on both sides, as in (1)

Note that AdS_2 has an $SL(2)$ isometry group. However, the group of asymptotic symmetries is larger; it is the full group of time reparametrizations. It turns out that in the zero energy limit these are indeed symmetries of the correlators. In other words, the Schwarzian mode comes from the spontaneous breaking of this symmetry, and its action from the explicit breaking [17, 22]. This action becomes irrelevant at low energies and the integral over this mode restores the symmetry. The integral is finite thanks to supersymmetry.

2.5 Lorentzian continuation

These results imply that Lorentzian correlators (anchored near the $NAdS_2$ boundary) also go to a constant at long times. In particular the two point function goes to a constant. It is important that this constant is real. This means that if one perturbs the black hole in a physical way, via a unitary process, then the effects of the perturbation will die out at long times. This is true because the change in correlation functions due to the perturbation is proportional to the commutator with the operator performing the computation. This commutator involves the imaginary part of the correlator, which vanishes at long times [1]. Notice also that the fact that the late-time Lorentzian 2-pt function is the same as the Euclidean 2-pt function implies that the bulk proper time between two points on the boundary is also finite, of order $2 \log S_e$.

If we perturb the extremal black hole by adding a simple UV operator localized in time, then we would raise its energy above extremality. We can avoid this by integrating the operator over a long Lorentzian time, a time longer than $1/E_{\text{gap}}$. This has the effect of projecting the operator to the zero energy subsector. We expect that these operators are similar to the operators $\hat{O} = POP$ that we were discussing above in the Euclidean context. We will make this approximation when we discuss probing black holes in the next subsection.

2.6 Building and exploring wormholes

We have already mentioned that the state (7) corresponds to a supersymmetric wormhole of finite length. This wormhole is empty, it contains no bulk particles.

We now consider adding an operator during the Euclidean evolution that produces the wormhole state, see figure 6a. This produces a particle in the middle of the wormhole. There is a unique state

⁶The actual propagator depends on more than just the distance through some extra factors that, when inserted in (19), also decay at long distances.

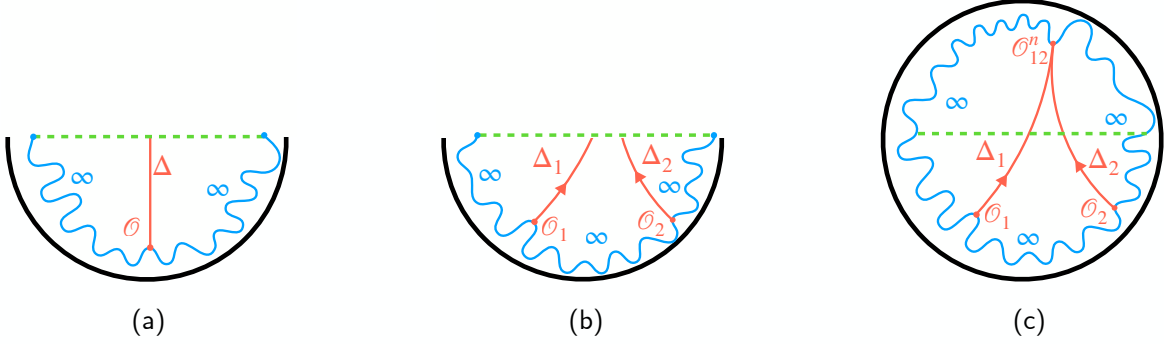


Figure 6: (a) State created by the insertion of an operator. It contains matter on the wormhole (indicated by green). (b) State created by the insertion of two operators, now we have two particles in the wormhole. (c) Overlap between the state in (b) and a state like (a) but created with the two particle primary operator O_{12}^n .

that we get by acting with a single conformal primary operator in this fashion⁷. For this state we expect that the distance between the two boundaries becomes bigger than for the empty wormhole. One way to estimate this distance is by computing an out of time order correlator involving the operator we inserted, which has dimension Δ and another operator with dimension Δ' . As $\Delta' \rightarrow 0$ we are measuring the length between the two boundaries, see figure 7. Using essentially this method we find the the distance is [1]

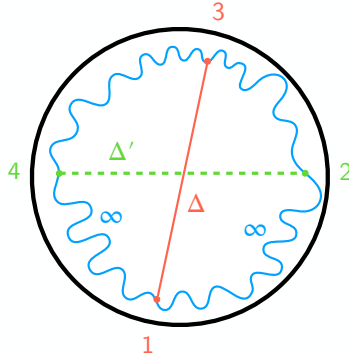


Figure 7: We consider a 4pt function involving a pair of heavy operator of dimension $\Delta \gg 1$ and a very light operator between points 2 and 4. If we do not insert any operator at 2 and 4, we can view it as giving a wormhole going between 2 and 4. This is a wormhole that contains matter in the representation with weight Δ . The typical distance between 2 and 4 gives us an estimate for the size of the wormhole.

$$\ell_{24} = 2 \log \Delta + o(1) , \quad \text{for } \Delta \gg 1 \quad (20)$$

which is saying that, as expected, the distance between 2 and 4 is getting larger than the empty wormhole (11).

The states we are considering would have the following interpretation in a quantum mechanical dual. The thermofield double is related to the operator P that projects on to the ground states,

⁷By conformal primary, we mean the boundary operator that behaves as an $SL(2)$ primary in the finite temperature conformal regime of $NAdS_2/NCFT_1$.

viewed as an entangled state. As a state on the zero energy subspace this is the identity and has maximal entropy. The state with a particle is given by

$$\hat{O} = POP \quad (21)$$

Since it is a deformation of the maximal entropy state, this state would have less entropy. The entropy can be computed via the replica trick, which involves computing $2n$ correlation functions. In general these are complicated. However, if Δ is large we expect that the OTOC contributions are suppressed, because of the increased distance we mentioned in (20). Then only planar diagrams contribute and we get an entropy that is lower than maximal by an amount that is Δ independent, for large Δ [1].

The same computation enables us to calculate the distribution of eigenvalues of the density matrix, which is simply related to the distribution of eigenvalues of POP . The latter turns out to be that of a gaussian random matrix, given by a semicircle law [1]. The connection between bulk fields and random matrices was made previously, and in more generality, in [23], generalizing the discussion of pure JT gravity [19].

Note that we expect that any state that contains matter will typically have lower entropy than the empty wormhole. This is particularly the case for states generated by the insertion of operators through Euclidean evolution, as we described above. The only exception would be states that are obtained by the action of a unitary operator on the low energy states. This can be achieved by performing a very slow Lorentzian evolution.

Another interesting question is the following. Imagine that now we add two particles that are well separated in the Euclidean evolution so that we get the state given by

$$\hat{O}_1 \hat{O}_2 = PO_1 PO_2 P \quad (22)$$

This is expected to be a wormhole that contains a pair of particles, see figure 6b. The two particle state can be decomposed into a set of representations of $SL(2)$ with dimensions $\Delta_1 + \Delta_2 + n$, which we denote as O_{12}^n . Each of these representations gives rise to a single wormhole state, so that the full two particle wormhole is a superposition of wormholes each associated to a value of n , $PO_{12}^n P$. We can find the amplitude for each n by computing the overlap, see figure 6c,

$$\langle \hat{O}_{12}^n \hat{O}_1 \hat{O}_2 \rangle \propto \text{Tr}[PO_{12}^n PO_1 PO_2] \quad (23)$$

These overlaps seem to be finite and of order one, if n is not large. We interpret this as saying that the two particles present in the state (22) are not very far from each other. It would be interesting to pursue this further and establish more clearly the nature of the state.

Note that the states we have discussed above, the wormhole and the wormhole plus extra matter, constitute particular entangled BPS states of two black holes, see figure 6a,b. These states are not represented by geometries that are locally supersymmetric in the interior, because the matter that we insert can break supersymmetry. However, after we include the boundary mode and project to low energies, we end up with a state that is BPS. So this is a novel way to construct BPS states and it differs from the traditional construction of BPS gravity solutions which involve finding a background with Killing spinors [24]. As an analogy, imagine we want to find a state with angular momentum zero for a rigid body. We could consider excitations on the rigid body that are not rotational invariant but we can then adjust the overall rotation of the body to produce a zero angular momentum state. Note that the two black holes that we are considering here are in different universes. If they were in the same universe, then we could wonder whether they preserve a common supersymmetry. If they are far away in flat space, they would preserve different



Figure 8: (a) zero energy empty wormhole has a distance $2\tilde{d}$ (26) between the two sides. (b) We claim that the results on correlation function imply that microstates that are designed to be maximally different for a given operator O of order one conformal dimension should start to differ from each other only at a distance $\tilde{d}$ from the boundary.

poincare supersymmetries since they have opposite charges. It might be possible to embed them into an ambient space with suitable fluxes so that oppositely charged black holes end up being supersymmetric.

It appears that we could get an infinite amount of different states by adding various sequences of matter particles. However, we expect that the island formation phenomenon [25, 26, 27, 28] will imply that we cannot have more entanglement entropy than $2S_0$. Indeed, the finiteness of the cylinder 2-pt function is evidence of this claim, see [29]. In this paper, we restrict to a relatively low number of particles so that we do not need to worry about this.

2.7 Implications for the black hole microstates

In this section we make some comments on the interpretation of the long time two point function (13), which we reproduce here for convenience

$$e^{-S_0} \langle \hat{W} \hat{W} \rangle_\infty = \frac{1}{Z} \text{Tr}[PWPW] = c_\Delta^2, \quad c_\Delta^2 = \epsilon^{2\Delta} \frac{\Gamma(1+2\Delta)}{2^{4\Delta}} \propto \frac{1}{S_e^{2\Delta}} \quad (24)$$

We discuss implications of the c_Δ factor. It sets the scale of the size of the operator in the IR compared to the operator in the UV. c_Δ is also telling us about the typical size of the eigenvalues of the operator $\hat{W}$

$$e^{-S_0} \langle \hat{W} \hat{W} \rangle_\infty = e^{-S_0} \sum_{i=1}^{e^{S_0}} w_i^2 \quad (25)$$

where w_i are the eigenvalues of $\hat{W}$. So the typical eigenvalue is of order c_Δ (24).

It is important that they are suppressed by the factor $S_e^{-\Delta}$. This factor is present for all correlators and it arises from the propagation of the field from the boundary to a distance

$$\tilde{d} = \log S_e \quad (26)$$

into the interior, we interpret this as saying that all microstates have to agree with the AdS_2 geometry at least up to this distance into the interior, see figure 8.

We can call the states that diagonalize this operator, “fuzzball states”, in the sense that they maximize the difference in expectation value of the operator $\hat{W}$ from state to state. We use this name because “fuzzballs” are a hypothetical representation of the black hole microstates in terms of gravity solutions, and presumably such states are designed to maximize their differences as seen

by simple gravity operators. Here we are not arguing for or against fuzzball proposals, see [30]. We are only providing constraints that those proposals should obey if we want to interpret the AdS_2 geometries as arising from some statistical average over such states. Since $\hat{W}$ is a gravity mode, then these results constrain the form of these gravity modes for typical solutions. Furthermore, since we expect a factor of $e^{-\Delta_i \tilde{d}}$ for each operator, this also suggests that all solutions should be similar to each other up to a distance $\tilde{d}$ from the boundary. Notice that since $\tilde{d}$ is also the distance at which it is important to consider the quantum dynamics of the boundary mode, any fuzzball proposal needs to incorporate the quantum mechanics of this mode, which is what we described in this paper. Note that the extremal black hole case is a favorable one to understand the explicit gravity description of microstates because there is no boundary time dependence.

In [1], we argue that $\hat{W}$ behaves like a random matrix with a bounded spectrum. By computing the bounds on the spectrum, one can constrain the expectation value of $\hat{W}$ in any putative fuzzball state. We expect that for $\Delta \gtrsim 1$, the maximum eigenvalue of $\hat{W}$ satisfies $|\lambda| \leq c_\Delta \times O(1)$, although we did not compute the $O(1)$ constant except in the $\Delta \gg 1$ limit where $|\lambda| \leq 2c_\Delta$.

Note that even though the vev of $\hat{W}$ in a typical eigenstate of $\hat{W}$ is c_Δ , the vev of $\hat{W}$ in a general typical quantum state is much smaller. This is because a general typical state is a generic linear combination of the operators that diagonalize $\hat{O}$. In fact, we are assuming that the expectation value of $\hat{O}$ is zero, $\text{Tr}[\hat{O}] = 0$. Its expectation value could vary from state to state, but its average should obey [31]

$$\int d\mu_\psi \left[\langle \psi | \hat{O} | \psi \rangle \right]^2 = e^{-2S_0} \text{Tr}[\hat{O}^2] = e^{-S_0} c_\Delta^2 \quad (27)$$

which is smaller, by a factor of e^{-S_0} than the normalized disk two point function in (24). In fact, this extra factor of e^{-S_0} suggest that the cylinder diagram might be relevant. The cylinder two point function is supposed to arise from an average of couplings of something of the form [32]

$$e^{-2S_0} \overline{\text{Tr}[\hat{O}] \text{Tr}[\hat{O}]} = e^{-2S_0} \overline{\text{Tr}[OP] \text{Tr}[OP]} = \overline{\left(\frac{\text{Tr}[OP]}{Z} \right)} \overline{\left(\frac{\text{Tr}[OP]}{Z} \right)} = e^{-2S_0} \langle 2 \text{ pt} \rangle_{cyl} \quad (28)$$

where we introduced factors of e^{-S_0} so that we normalize the expectation values of $\hat{O}$ in the standard way. The couplings determine the form of the projector from the UV to the IR and therefore the form of the operator $\hat{O}$. There we expect that an average over couplings should have similar effects as the average over states for fixed couplings. More precisely, we have

$$\begin{aligned} e^{-2S_0} \overline{\text{Tr}[\hat{O}] \text{Tr}[\hat{O}]} &= e^{-2S_0} \sum_{i,k} \int d\mu_J \langle i | \hat{O} | i \rangle_J \langle k | \hat{O} | k \rangle_J \\ &\approx e^{-2S_0} \sum_i \int d\mu_J \left(\langle i | \hat{O} | i \rangle_J \right)^2 \approx e^{-S_0} \int d\mu_\psi \left[\langle \psi | \hat{O} | \psi \rangle \right]^2 \end{aligned} \quad (29)$$

where J denotes the couplings and $\int d\mu_J$ is the average over couplings. The IR basis elements $|i\rangle_J$ and $|k\rangle_J$ depend on the couplings. In going from the top line to the bottom line we assumed that the average over couplings would produce a δ_{ik} . We then interpreted the average over couplings together with the sum, $e^{-S_0} \sum_i$, as similar to an average over states as we had in (27).

Indeed we see that (28) is the same as (27) once we use that (28) is given by the cylinder diagram (16), multiplied by e^{-2S_0} . This explains why we get the same value for the cylinder (16) and disk (10) two point functions, up to the expected factor of e^{S_0} . There is a similar relation at non-zero energies in the microcanonical ensemble, see appendix A.

The fact that the R charge of the vacuum is zero has an interesting implication. It means that any operator with non-zero R charge should be trivial on the BPS ground states. In fact,